Building DFA

Typical questions

Define a DFA which recognizes the given language L.

or

Prove that the (given) language L is regular.

Building DFA

Example

$\Sigma = \{0, 1\}$

$\delta \longrightarrow \underline{Q} = \{q_1, q_2\}$

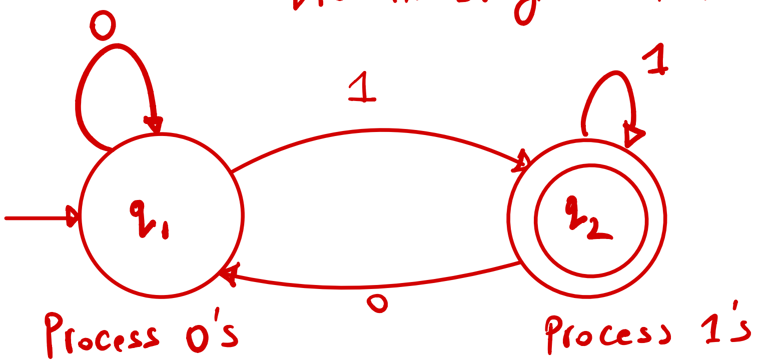
Define a DFA which recognizes

$$L_1 = \{w \mid w \text{ ends with a } 1\}$$

$L_1 = \{\underline{1}, 01, \underline{001}, 0101, \ldots\}$

$010 \notin L_1$

↳ For all strings in this set Machine says yes.



Input	Output
$q_1, 0$	q_1
$q_1, 1$	q_2
$q_2, 0$	q_1
$q_2, 1$	

Building DFA

Example

Define a DFA which recognizes

$$L_2 = \{w \mid w \text{ has substring } 11\}$$

Building DFA

0 1 2 3 4 5 6 7 8

even length: length 0, length 2, length 4

Example

Define a DFA which recognizes

$\Sigma = \{0, 1\}$

$\{\} = \phi \neq \{\varepsilon\}$

$L_3 = \{w \mid w \text{ has even length}\}$

$L_3 = \{00, 01, 0011, 0101, 111010, \varepsilon \text{ ---}\}$

0/1 0/1 0/1 0/1

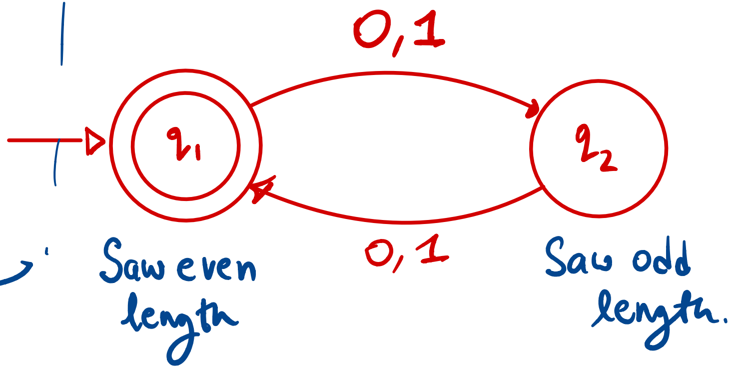
1st 2nd 3rd 4th

odd

even

odd

even



$q_1, 0 \rightarrow q_2$

$q_1, 1 \rightarrow q_2$

$q_2, 0 \rightarrow q_1$

$q_2, 1 \rightarrow q_1$

Building DFA

Example

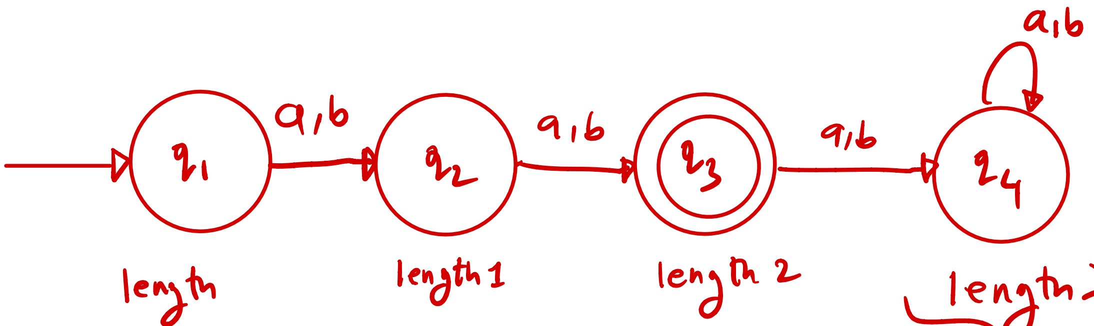
$\Sigma = \{a, b\}$

Define a DFA which recognizes

L_4 = {w | length of w is two}

$L_4 = \{aa, ab, ba, bb\}$

length 1 $\notin L_4$
Length 3 $\notin L_4$ } No
Length 2 $\in L_4$ } yes.



abaa ✗

$\underbrace{\text{length} \geq 3}_{\text{trap state/dead state.}}$ = at least 3

Building DFA

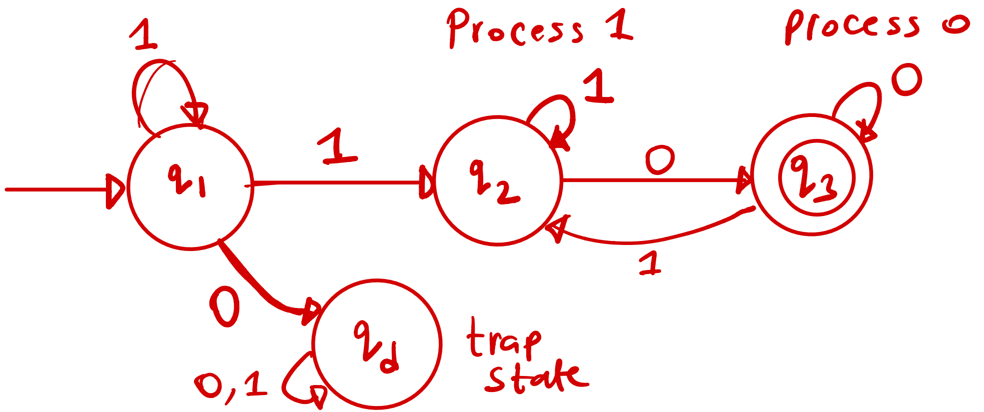
Example

Define a DFA which recognizes

$\Sigma = \{0, 1\}$

$$L_5 = \{w \mid w \text{ starts with 1 and ends with 0}\}$$

$L_5 = \{10, 110, 1001010, \_\ \_\}$



011 ✗

1101 ✗

Building DFA

Example

Define a DFA which recognizes

$\Sigma = \{a, b\}$

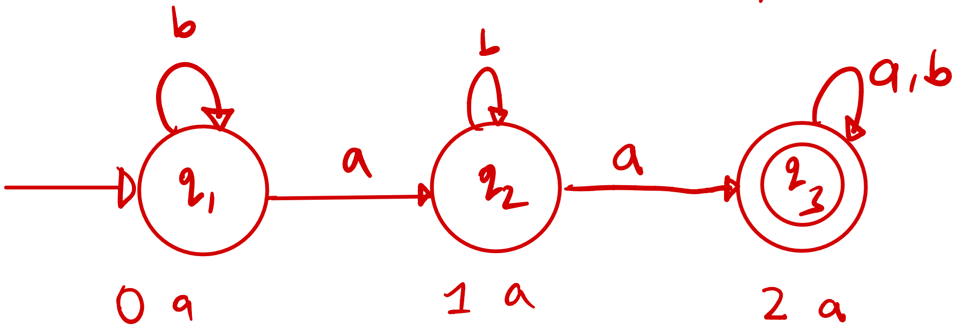
$$L_6 = \{w \mid w \text{ has at least 2 a's}\} \geq 2$$

2 a's

3 a's

n a's

$L_6 = \{aab, baa, aba, aa, aaab, ababa \ldots\}$



Building DFA

≤ 2

$\#a = 0$
$\#a = 1$
$\#a = 2$ } yes

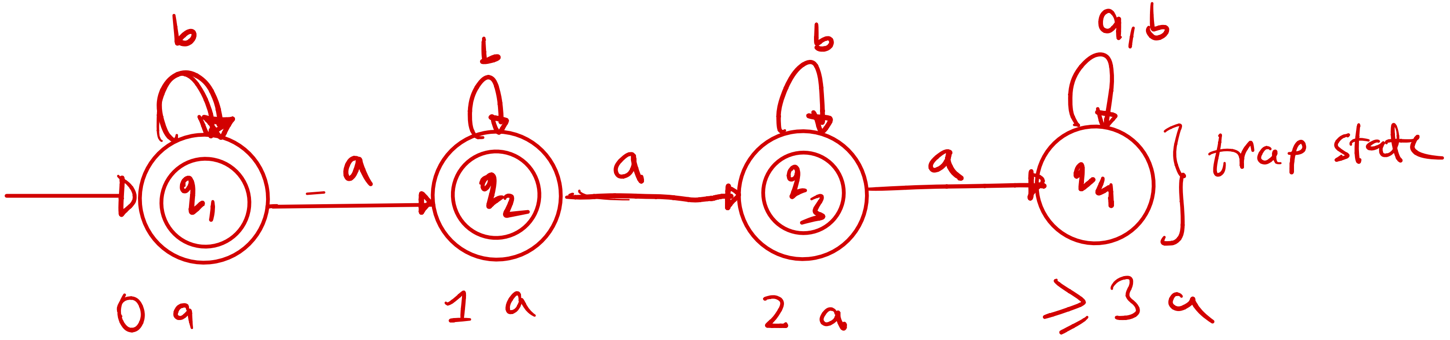
$a = 3$
$\vdots$
$a = n$ } No

Example

Define a DFA which recognizes

$$L_7 = \{w \mid w \text{ has at most 2 a's}\}$$

$L_7 = \{ \underline{bbb}, \underline{ba}, \underline{aab}$ __ __ $\}$



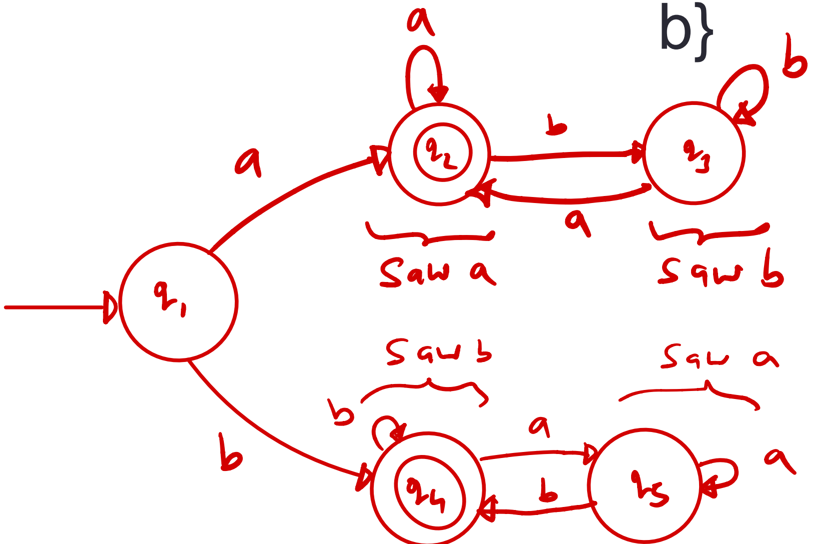
Building DFA

$L_8 = \{a, b, ababa, babab \_ \_ \_\}$

Example

Define a DFA which recognizes $\Sigma = \{a, b\}$

L_8 = {w | w starts and ends with a or w starts and ends with b}



Building DFA

No real Memory like RAM

$\Sigma = \{0,1\}$

3 bit memory

$= 2^3 = 8$

Remember

$000 = 0$
$001 = 1$
$010 = 2$
$011 = 3$
$100 = 4$
$101 = 5$
$110 = 6$
$111 = 7$

8 state

States are our only (computer) memory.

Design and pick states with specific roles / tasks in mind.

"Have not seen any of desired pattern yet"

"Trap state"

Justification?

To prove that the DFA we build, M, actually recognizes the language L

DFA

$L(M) = 2, 3, 4$

$L = 2, 3, 4, 5$

$L(M) \subseteq L$

WTS $\underbrace{L(M)}_{\text{Machine language}} = \underbrace{L}_{\text{Given language}}$

every element of $L(M)$ contained in L.

(1) Is every string accepted by M in L? $L(M) \subseteq L$

(2) Is every string from L accepted by M? $L \subseteq L(M)$

or contrapositive version: Is every string rejected by M not in L.

Set $L(M)$, L

$L(M) \subseteq L \wedge L \subseteq L(M) \equiv L(M) = L$